

Sub-Surface Remediation & Joint Sovereignty Protocol

"The Nuclear Dust Protocol"

Self-Contained · Incorporable by Reference · Step-Back/Earnback Integrated

Version 2.5 — Integrated Stabilization Measure

Expires automatically upon completion of remediation

or invocation of Article 4 entombment protocol

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Preamble

The Parties acknowledge that the relevant highly enriched uranium (HEU) exists no longer in intact form but as dispersed particulate matter ("Nuclear Dust") beneath collapsed geological strata at previously declared military engineering sites.

No Party accepts unilateral armed foreign excavation on its sovereign territory.

A verified, multilateral extraction mechanism recovers the Nuclear Dust to IAEA-certified standards while preserving Iranian surface sovereignty and excluding active-duty U.S. and Iranian military personnel from the remediation site.

The Parties adopt this Annex V as the binding operational mechanism governing extraction, verification, and financial sequencing for all Nuclear Dust remediation activities. This Annex treats every obligation as a live operational trigger, not a future promise.

Present-Tense Execution Principle

Every clause in this Annex executes in the active present tense. Consequences activate in real time. No Party waits for subsequent authorization once a condition is met. Triggers fire automatically. Penalties accrue immediately. Releases occur upon verification, not upon political consensus.

Step-Back Integration Principle

This Annex V incorporates the full Three-Tier Breach & Step-Back Architecture from the parent framework's Article 13, calibrated specifically for Nuclear Dust remediation operations. The tier system governs all micro-tranche releases, penalty assessments, and earnback procedures. No violation triggers automatic military escalation. The framework architecture excludes kinetic enforcement mechanisms.

The Immutable Present-Tense Matrix (V2.5)

This Annex V operates under the Immutable Present-Tense Matrix V2.5. All clause construction adheres to the following governing constraints: The framework architecture excludes the vocabulary items "shall," "must," "can," "may," "should," and "will" from all operational provisions. Every

sentence carries a clear, active subject performing a direct, present-tense action. The framework architecture excludes passive constructions, hierarchical ordering phrases, and unilateral restraint language. All automated consequences activate as immediate, real-time operational realities. Every prohibitive commitment operates as a feature of the shared framework architecture, not as a concession or voluntary limitation by any single Party.

Modularity Clause. This Annex V is self-contained and is incorporable into any existing or future agreement between the Parties through a single sentence of adoption. It does not amend or supersede conflicting provisions of any parent framework unless the Parties explicitly so agree. In the event of inconsistency, this Annex V governs with respect to Nuclear Dust remediation, dual-key sovereignty, financial micro-tranche sequencing, CANDOR governance, three-tier breach and step-back protocols, emergency break-point protocols, and earnback architecture.

Article 1: Coordinated Access for Nuclear Dust Remediation (CANDOR)

1.1 CANDOR Composition and Governance

The Coordinated Access for Nuclear Dust Remediation consortium ("CANDOR") operates as the sole authorized international body for Nuclear Dust extraction. CANDOR comprises the following entities with defined roles:

Table V-1 CANDOR Composition and Roles

Entity	Role	Responsibilities
People's Republic of China	Co-Lead (Operational)	Site command coordination; extraction engineering oversight; dispute resolution between CANDOR members; liaison to Tehran and Washington
International Atomic Energy Agency (IAEA)	Co-Lead (Verification)	Radiological assay and mass balance certification; containerization standards enforcement; seal application and verification; real-time data feeds to all Parties
European Signatories (E3+: UK, France, Germany, EU representative)	Technical Operations	Specialized extraction equipment provision; environmental monitoring; waste containment engineering; technical training for non-military excavation crews
Russian Federation	Observer / Escrow Guarantor	Independent observation of CANDOR operations; escrow fund oversight for financial tranche releases; technical advisory on sub-surface recovery methodology

1.2 Personnel Exclusions and Access Rules

The following personnel restrictions apply at all CANDOR operational sites:

- No active-duty U.S. military personnel enters Iranian territory for CANDOR purposes. U.S. engineering expertise reaches the site only through third-party private contractors operating under IAEA badge authority and live video feed access for both Washington and Tehran.

- b. No active-duty Iranian military personnel enters the CANDOR operational perimeter during active extraction. Iranian Atomic Energy Organization (AEOI) technical observers maintain a presence in the site coordination center outside the active extraction zone.
- c. All CANDOR personnel carry IAEA-issued credentials. Iran's Ministry of Foreign Affairs receives the full personnel roster 72 hours before each rotation. Iran retains the right to deny entry to any individual on security grounds, with written justification provided within 24 hours.
- d. Maximum CANDOR non-Iranian personnel at any given time: 85 individuals. This cap includes Chinese operational staff, European technical personnel, IAEA inspectors, and Russian observers. The cap excludes AEOI technical observers and Iranian site security personnel.

1.3 Extraction Mandate and Standards

CANDOR physically excavates, recovers, and containerizes a minimum of 99.7% of the estimated 440.9 kg of 60%-enriched HEU existing as Nuclear Dust. The IAEA certifies recovery percentage through mass spectrometry and radiological assay at each extraction milestone.

Table V-2 Recovery Milestones and Verification Standards

Milestone	Recovery Target	IAEA Verification Method	Timeline
Initial Site Survey	Confirm dust presence and distribution	Core drilling assay; gamma spectroscopy	Days 1–14
Phase I Extraction	25% recovery (110.2 kg)	Real-time mass balance; container seal verification	Days 15–60
Phase II Extraction	75% recovery (330.7 kg cumulative)	Continuous assay; environmental contamination monitoring	Days 61–120
Final Containerization	99.7% recovery (439.6 kg cumulative)	Final mass spectrometry; container certification; chain-of-custody documentation	Days 121–180

1.4 Operational Security Protocols

CANDOR operates under the following security framework:

- A 5-kilometer exclusion zone surrounds each operational site. Iranian police maintain the outer perimeter. CANDOR security maintains the inner perimeter.
- All extraction equipment enters through a single designated logistics corridor, inspected by joint Iranian-IAEA teams.

- Every container of recovered Nuclear Dust receives an IAEA tamper-evident seal with unique serial number at the point of extraction.
- Live encrypted video feeds transmit to monitoring stations in Tehran, Washington, Beijing, Vienna (IAEA), and Moscow simultaneously. No feed interruption exceeds 15 minutes without automatic escalation to the Joint Secretariat.

Article 2: Dual-Key Sovereignty Provisions

2.1 Digital Key Architecture

Each excavation shift requires two digital authorization keys operating in tandem:

- a. **Iran Key:** The AEOI Director General (or designated deputy) holds one digital key. This key authorizes the shift from the Iranian sovereignty perspective. Without this key, no extraction equipment activates.
- b. **IAEA Key:** The IAEA Director General (or designated senior inspector on-site) holds the second digital key. This key certifies that radiological safety conditions, extraction protocols, and verification standards are met. Without this key, no extraction equipment activates.

Both keys operate simultaneously for every shift. Either key holder withdraws authorization at any time, halting extraction immediately. A key withdrawal triggers automatic notification to all Parties within 15 minutes via the Shared Digital Registry.

2.2 U.S. Personnel Access Restrictions

The following access rules apply specifically to U.S. nationals:

- a. No U.S. national enters Iranian territory for CANDOR purposes without explicit, time-limited AEOI visa approval. Each visa is valid for a single shift rotation (maximum 12 hours) and a specific defined zone.
- b. The total number of U.S. nationals present at any given time does not exceed 30 individuals. All 30 are non-uniformed civilian contractors operating under IAEA badge authority.
- c. U.S. nationals perform no operational role in direct extraction. Their permitted activities include: remote engineering consultation via secure video link; technical data review in the CANDOR coordination center; and observation of IAEA verification procedures. Any U.S. remote consultation that results in a change to extraction methodology or equipment configuration requires prior written approval from the CANDOR Chinese co-lead and IAEA on-site verification lead, recorded in the Shared Digital Registry within one hour.
- d. Iran receives the full name, passport number, and professional biography of each U.S. national 7 days before visa issuance. Iran approves or denies each application within 72 hours. A denial requires written justification; a failure to respond within 72 hours constitutes automatic approval.

2.3 Iranian Surface Sovereignty

Iran retains full surface sovereignty over all CANDOR operational sites. This sovereignty includes:

- Ultimate authority over all personnel entering Iranian territory for CANDOR purposes
- Control of the outer security perimeter and all access roads to the operational sites
- The right to suspend CANDOR operations for up to 48 hours in response to verified security threats, with automatic notification to all Parties and the Joint Secretariat
- Jurisdiction over any criminal or civil matters arising outside the CANDOR operational perimeter

2.4 Easement Terms and Revocation

CANDOR operates under a limited, revocable easement for access to the three defined collapsed zones. Annex V-2A contains the GPS-locked coordinates of these zones.

The easement terms are:

Scope: The easement permits only sub-surface extraction, containerization, and environmental remediation activities. It does not authorize permanent construction, military presence, or infrastructure development unrelated to Nuclear Dust recovery.

Duration: The easement remains valid for the duration of CANDOR active operations, up to a maximum of 180 days from the date of Annex V entry into force.

Revocation: Iran revokes the easement with 72 hours' written notice to the Joint Secretariat and all CANDOR members. Revocation takes effect strictly at the conclusion of the current shift; no revocation terminates an active extraction shift mid-cycle. Upon revocation, Article 5 (Emergency Break Points) governs the financial and operational transition.

Renewal: The Parties extend the easement by mutual written agreement. No single Party extends the easement unilaterally.

Article 3: Financial Micro-Tranche Lockstep ("Dust-for-Dollars")

3.1 General Architecture

The disputed \$12 billion in frozen Iranian assets held in escrow is not released upfront. Instead, asset release occurs in strict micro-tranches tied to verified extraction milestones. The Swiss facilitator administers the escrow account and executes releases within 24 hours of IAEA certification.

The following framework architecture governs all micro-tranche releases:

- IAEA certification constitutes the sole release trigger. No political entity overrides an IAEA-certified release.
- Every release generates an automatic entry in the Shared Digital Registry (Annex U, Article 5), timestamped and visible to all Parties.
- The Three-Tier Breach & Step-Back Architecture under Article 4 of this Annex governs all penalty assessments, benefit reductions, and earnback procedures. No release occurs during an active Tier 2 or Tier 3 remediation window unless IAEA clean certification activates the release.
- Humanitarian tranches enjoy protected status. The framework architecture excludes humanitarian funds from any step-back reduction, regardless of tier classification.

3.2 Initial Assay Tranche

Upon IAEA verification of Nuclear Dust presence through on-site radiological assay, the Swiss facilitator releases \$500 million from the frozen escrow. This release occurs within 24 hours of IAEA certification. The assay verifies that a minimum of 400 kg of HEU-equivalent dispersed material is present at the designated sites.

The Humanitarian Firewall protects this initial tranche. No tier breach — including Tier 3 — triggers rescission, clawback, or hold of funds already released through prior tranches. The "no clawback" provision is absolute and non-waivable.

3.3 Per-Ton Extraction Tranches

For every metric ton of Nuclear Dust extracted (measured in 10-ton cumulative tranches), the Swiss facilitator releases \$200 million from the escrow. The release triggers are:

Table V-3 Per-Ton Extraction Tranche Schedule

Cumulative Extraction	Tranche Release	Cumulative Released	Verification Trigger
10 tons	\$200M	\$700M (\$500M + \$200M)	IAEA mass balance confirms 10 tons containerized
20 tons	\$200M	\$900M	IAEA mass balance confirms 20 tons containerized
30 tons	\$200M	\$1.1B	IAEA mass balance confirms 30 tons containerized
40 tons	\$200M	\$1.3B	IAEA mass balance confirms 40 tons containerized
50 tons	\$200M	\$1.5B	IAEA mass balance confirms 50 tons containerized
60 tons	\$200M	\$1.7B	IAEA mass balance confirms 60 tons containerized
70 tons	\$200M	\$1.9B	IAEA mass balance confirms 70 tons containerized
80 tons	\$200M	\$2.1B	IAEA mass balance confirms 80 tons containerized
90 tons	\$200M	\$2.3B	IAEA mass balance confirms 90 tons containerized
100 tons	\$200M	\$2.5B	IAEA mass balance confirms 100 tons containerized

Note: The 100-ton figure represents total excavated material including soil matrix and debris. CANDOR recovers the 440.9 kg HEU equivalent from within this matrix. The IAEA certifies both total tonnage and HEU concentration per tranche.

3.4 Final Containerization Payment

Upon IAEA certification that containerization of 99.7% of the Nuclear Dust is complete and the material is ready for escorted transport to neutral-country storage, the Swiss facilitator releases the remaining balance of the \$12 billion escrow, minus the amounts already released through prior tranches.

In addition to the remaining balance, the Swiss facilitator disburses a \$1 billion completion bonus, which splits 50/50 between:

- Iran's Central Bank (50% = \$500 million), deposited as sovereign liquid funds with no restriction; and
- A designated humanitarian escrow (50% = \$500 million), administered by UN OCHA for food and medicine procurement benefiting the civilian population of conflict-affected regions in the Middle East, with disbursement priorities determined jointly by Iran's Central Bank and UN OCHA, subject to neutral third-party audit.

3.5 Escrow and Verification Mechanism

The Swiss facilitator maintains the escrow account with the following operational rules:

- The Central Bank of Oman hosts the escrow account under sovereign immunity protections.
- IAEA certification is the sole release trigger. No political entity overrides an IAEA-certified release.
- Every release generates an automatic entry in the Shared Digital Registry (Annex U, Article 5), timestamped and visible to all Parties.
- If technical review delays IAEA certification beyond 72 hours, the Swiss facilitator holds the funds in a pending-release sub-account earning interest at the Omani central bank rate.
- Russia, as escrow guarantor, receives real-time notification of every deposit, release, and balance change. Russia does not hold a release key but serves as an independent balance auditor.
- If IAEA certification is withheld due to an active tier breach remediation window, the corresponding tranche remains in pending status. The tranche releases automatically upon IAEA clean certification and JIG confirmation of remediation completion.
- No interest accrues to either Party during an active Tier 3 remediation period.

Article 4: Three-Tier Breach & Step-Back Architecture

This Article integrates the parent framework's Human-in-the-Loop Governance with proportional step-back and earnback mechanisms, calibrated specifically for Nuclear Dust remediation operations. The framework architecture excludes kinetic enforcement mechanisms. Every penalty is financial and diplomatic, not military. Step-back informs; humans decide.

4.1 Tier 1: Technical Dispute (De Minimis)

Classification: Technical Dispute — De Minimis Breach

Telemetry Signature: Enrichment or extraction metrics register less than 0.5% above certified ceilings; no more than 2 un-mothballed rotors detected; IAEA reporting delays do not exceed 48 hours; CANDOR personnel count exceeds the 85-person cap by no more than 3 individuals; dual-key authorization delays do not exceed 2 hours per shift.

Base Response — Tier 1 Step-Down:

- No benefit reduction occurs. No tranche hold activates.
- A 14-day resolution window opens automatically.
- The Joint Implementation Group (JIG) provides technical assistance.
- Quiet compliance corrections remain confidential.
- No entry appears on the public compliance dashboard.

Tier 1 Enrichment Ceiling (Analogous Application): For CANDOR operations, the Tier 1 threshold applies to: extraction methodology deviations of less than 0.5% from certified protocols; environmental contamination readings marginally above baseline; container seal administrative delays of up to 48 hours; and mass balance discrepancies of less than 0.3% attributable to measurement tolerance.

4.2 Tier 2: Material Disruption

Classification: Material Disruption — Significant Operational Breach

Telemetry Signature: Extraction metrics register more than 0.5% above certified ceilings; more than 2 un-mothballed rotors detected; IAEA reporting delays exceed 72 hours; CANDOR personnel count exceeds the 85-person cap by more than 3 individuals; unauthorized equipment enters the logistics corridor; active-duty military personnel from any Party enter restricted zones; tamper-evident seal integrity compromise detected; or extraction operations proceed without dual-key authorization.

Base Response — Tier 2 Step-Down:

- A base 25% reduction in non-humanitarian benefits activates automatically.
- Humanitarian tranches remain protected. The framework architecture excludes humanitarian funds from reduction.
- The SWIFT phase holds for exactly 14 days as a remediation fuse.
- A 30-day remediation window opens.
- Full restoration occurs on IAEA clean certification within 14 days.
- The affected Party receives JIG technical assistance throughout the remediation period.

Proportional Tier 2 Penalty (Short-Duration Reduction): A Tier 2 violation lasting less than 7 days triggers a 10% reduction, not 25%. The full 25% reduction activates only after the 30-day remediation window expires without compliance restoration. This proportionality rule prevents minor or quickly-cured violations from triggering disproportionate penalties.

4.3 Tier 3: Deliberate Hostile Violation

Classification: Deliberate Hostile Violation — Existential Threat to the Framework

Telemetry Signature: Evidence of nuclear weapons activity; fissile material diversion; IAEA attack or obstruction; deliberate extraction of material outside IAEA verification; breach of tamper-evident seals with apparent intent; unauthorized transport of Nuclear Dust outside designated corridors; destruction of monitoring infrastructure with apparent intent; or military incursion into the CANDOR operational perimeter by any Party.

Base Response — Tier 3 Step-Down:

- A base 50% reduction in non-humanitarian benefits activates automatically.
- Tier 3 modules enter hold status. New tranche releases pause.
- A 90-day remediation window opens.
- Full earn-back requires: 90 consecutive days of sustained compliance plus IAEA clean certification plus JIG majority confirmation.
- During the remediation period, no interest accrues to either Party on held funds.
- Humanitarian tranches remain protected. The framework architecture excludes humanitarian funds from reduction.

Proportional Tier 3 Penalty (Immediate Remediation Reduction): A Tier 3 violation that is immediately remediated (within 72 hours of detection) triggers a 25% reduction, not 50%. The full 50% reduction

activates only after the 90-day remediation window expires without sustained compliance restoration. This proportionality rule rewards immediate corrective action.

4.4 Proportional Penalty Calibration

Table V-4 Proportional Penalty Schedule

Tier	Duration/Condition	Penalty	Remediation Window
Tier 2	Less than 7 days	10% reduction	30 days
Tier 2	7 days or more / after remediation expiry	25% reduction (full)	30 days
Tier 3	Remediated within 72 hours	25% reduction	90 days
Tier 3	After 90-day remediation expiry	50% reduction (full)	90 days

4.5 The Earnback Protocol

The earnback protocol ensures that compliance restoration yields proportional benefit restoration. The framework architecture rewards sustained correction, not punishes past error indefinitely.

Tier 1 Earnback: Full benefits remain intact. The Parties need no earnback procedure. The 14-day resolution window closes upon JIG confirmation of remediation.

Tier 2 Earnback: Full benefit restoration occurs upon IAEA clean certification within the 30-day remediation window. The restoration activates automatically upon JIG certification of compliance. The compliance clock resets to 30 days (not zero) from the date of remediation for a single Tier 2 violation lasting less than 7 days. A second Tier 2 violation within the same 90-day period resets the compliance clock to zero. Any Tier 3 violation resets the compliance clock to zero.

Tier 3 Earnback: Full earn-back requires all three conditions: 90 consecutive days of sustained compliance with zero tolerance for additional violations; IAEA clean certification issued no earlier than Day 90 of the remediation period; and JIG majority confirmation (excluding the affected Party's JIG members) that compliance is genuine and sustained. Upon satisfaction of all three conditions, the Swiss facilitator releases all held tranches within 24 hours.

No Clawback of Released Funds: Funds already released from escrow prior to the activation of a step-back penalty are not subject to return, clawback, or rescission under any circumstances. The withdrawing Party retains full title and control over all previously released assets. The penalty affects only future tranches not yet released as of the step-back activation date. This "no clawback" provision is absolute and non-waivable.

4.6 Human-in-the-Loop Governance

No automated output triggers a binding diplomatic consequence without explicit human authorization at the appropriate level of seniority.

Dual-Key Authorization: Any transition that materially alters the obligations of either Party requires simultaneous authorization from designated senior officials on both sides. The JIG serves as the primary compliance assessment body and meets monthly by secure video. Within 72 hours of any reported discrepancy, the JIG convenes for emergency consultation.

Grace Over Rigidity: A Party that needs an extra 48 hours because its treasury office closes for a national holiday receives treatment as a partner whose administrative rhythms deserve accommodation, not as a violator. Administrative delays of up to 48 hours for verified logistical constraints do not constitute a material breach or trigger step-back.

Political Shock Absorber: Framework implementation enters automatic Review Pause (not termination) upon: elections; assassination of senior leadership; cabinet collapse; mass protests; leadership transitions; or kinetic disruptions within regional security architectures. During Review Pause: Tier 1-2 modules continue; Tier 3 modules enter hold; the Secretariat issues a Review Pause Certificate within 48 hours; the Parties have 30 days to confirm continuation, extendable by mutual agreement up to a maximum of 90 days. Review Pause does not constitute a material breach and does not trigger the three-tier system.

4.7 Attribution Deliberation Media Blackout

When the JIG conducts attribution deliberations for any incident that could trigger Tier 2 or Tier 3 step-back, a mandatory 72-hour media blackout applies to all Parties and guarantor states.

During this period:

- No Party issues official or on-record statements attributing responsibility for the incident.
- Leaks of attribution deliberations trigger investigation by the Joint Secretariat; responsible officials face Tier 1 technical dispute proceedings.
- The blackout lifts automatically upon JIG issuance of a preliminary attribution finding, or after 72 hours if no finding issues.
- The blackout does not apply to unclassified operational reporting (e.g., "an incident occurred at location X at time Y").

Senior Official Leak Escalation: The JIG receives referral of any deliberate leak attributable to a senior official (minister-level or above, or their immediate designee) for potential Tier 2 step-back consideration if the leak materially prejudices attribution deliberations. The JIG determines within 14 days whether the leak

was deliberate and materially harmful. If so, the violating Party faces a 5% benefit reduction in the affected module for 30 days, in addition to the Tier 1 demerit. Low-level administrative leaks remain at Tier 1. This provision is not a sunset; it constitutes a performance-based consequence for bad-faith conduct.

Article 5: Emergency Break Points

5.1 90% Recovery Threshold

If geological conditions, radiation levels, or structural collapse permanently block extraction below the 90% threshold, the Parties revert automatically to in-situ monitored entombment.

The determination requires:

- Unanimous certification by the CANDOR technical leadership (China co-lead, IAEA co-lead, and European technical lead)
- Independent verification by a second-party engineering assessment team (comprising non-CANDOR specialists from neutral countries, selected by the Joint Secretariat)
- Written confirmation that no alternative extraction methodology exists that is capable of achieving 90% recovery within the remaining operational window

5.2 In-Situ Entombment Protocol

Upon invocation of the 90% threshold break point, CANDOR transitions to the in-situ monitored entombment protocol:

- a. Permanent IAEA-approved containment barriers seal all remaining Nuclear Dust in place.
- b. The IAEA installs perpetual camera surveillance and continuous radiation monitoring at all entombment sites. Data feeds transmit in real time to Tehran, Washington, Beijing, Vienna, and Moscow.
- c. CANDOR applies tamper-evident seals at every access point. Any breach of these seals triggers automatic notification to all Parties within 5 minutes and constitutes a material breach of this Annex V.
- d. The IAEA conducts quarterly physical inspections of the entombment sites. Annual comprehensive reviews assess seal integrity, radiation levels, and structural stability.
- e. No further excavation occurs at the entombment sites without unanimous consent of all Parties and IAEA safety certification.

5.3 50% Release / 50% Hold Split

Upon invocation of the entombment protocol, the Swiss facilitator automatically executes the following split of the frozen \$12 billion:

50% Released (\$6 billion): Released to Iran's Central Bank within 48 hours of IAEA entombment certification. This amount compensates Iran for the verified extraction work completed and acknowledges the permanent IAEA monitoring commitment.

50% Held (\$6 billion): Remains in escrow until a joint U.S.-Iranian technical commission confirms entombment integrity. The commission mediates through China and comprises three U.S. technical experts, three Iranian technical experts, and one Chinese chair.

5.4 Joint Technical Commission Review

The joint U.S.-Iranian technical commission (mediated by China) convenes within 30 days of entombment protocol invocation. The commission executes the following mandate:

- a. Reviews all IAEA entombment data, seal integrity reports, and radiation monitoring logs
- b. Conducts an independent on-site inspection within 60 days of commission formation
- c. Issues a binding determination on entombment integrity within 90 days of commission formation
- d. If the commission confirms entombment integrity by majority vote (5 of 7 members), the Swiss facilitator releases the remaining \$6 billion to Iran's Central Bank within 24 hours
- e. If the commission determines that compromised entombment integrity exists, the Parties negotiate remedial measures. During remediation, the held \$6 billion remains in escrow. No interest accrues to either Party during the remediation period

Article 6: Verification and Telemetry Architecture

6.1 Real-Time Monitoring Infrastructure

CANDOR maintains the following real-time monitoring infrastructure at every operational site:

Table V-5 Telemetry Infrastructure and Data Feeds

System		Function	Feed Recipients			Latency	
Gamma Spectroscopy		Continuous radiological monitoring of extraction zone	IAEA Tehran;	Vienna; U.S. DOE	AEOI	Real-time (5-second refresh)	
Encrypted Network	Video	Visual monitoring of all extraction and containerization activities	All Parties; Secretariat		Joint	Live (15-second max interruption)	
Container Tracker	Seal	GPS location and integrity tracking of every sealed container	IAEA; Russian escrow	Swiss facilitator; auditor		1-minute updates	position
Environmental Sensor Grid		Air, soil, and groundwater contamination monitoring	Iran EPA; European technical team	IAEA;		15-minute cycles	sampling
Mass Ledger	Balance	Real-time tracking of extracted material weight and HEU concentration	All Parties; Shared Digital Registry			Updated per shift (every 12 hours)	

6.2 IAEA Mass Balance Verification

The IAEA certifies the mass balance for each extraction tranche through the following procedure:

- a. Pre-extraction baseline measurement of the extraction zone volume and estimated HEU dispersion
- b. Per-shift weighing of all extracted material with certified industrial scales (calibrated daily)
- c. Per-shift gamma spectroscopy assay to determine HEU concentration in the extracted matrix
- d. Post-shift reconciliation: total weight multiplied by HEU concentration equals HEU mass recovered
- e. Independent verification by a second IAEA inspector team using separate equipment
- f. Publication of the certified mass balance to the Shared Digital Registry within 6 hours of shift completion

6.3 Data Sharing Symmetry

All Parties share telemetry data symmetrically. No Party receives preferential access, earlier notification, or exclusive data feeds. Every data stream transmits simultaneously to all designated recipients. Any data asymmetry constitutes a Tier 2 Material Disruption under this Annex V.

Article 7: Implementation Timeline and Phases

The timeline balances the political demand for rapid, visible results against the physical reality of safe extraction operations. The framework structures the 180-day window around political momentum milestones that demonstrate progress while allowing the engineering teams the full operational duration they physically need to safely process hundreds of tons of contaminated soil matrix.

A spokesperson aggressively markets Day 14 (IAEA lock on the site, live feeds active, verification proven) and Day 30 (first physical tranches moving) as definitive proof of a hyper-accelerated, highly successful timeline. Meanwhile, the excavation crews receive the full 180 days they physically need to safely process the contaminated soil matrix without causing an engineering disaster.

7.1 Phase I: The Political & Political Momentum Window (Days 1–30)

Instead of pulling all material out of the ground in the first month, this window focuses on high-visibility, automated verification to give the White House an immediate win.

Days 1–7 — The Media Win

The CANDOR advance team arrives and equipment staging begins. The Encrypted Video Network installation provides live, 15-second latency feeds directly to Washington. This satisfies the "immediate transparency" requirement before a single grain of dust moves. The framework architecture treats live feed activation as the first operational milestone, not extraction volume.

Days 7–14 — The First Hard Metric

Core drilling assays and gamma spectroscopy confirm the exact location and distribution of the 440.9 kg of HEU. The IAEA issues its initial certification, triggering the release of the first \$500 million asset tranche. Within two weeks, the system proves reciprocity works. This constitutes the first hard verification milestone and activates the initial escrow release automatically.

Days 15–30 — The Compressed Demonstration

Active extraction begins in Zone 1. Crews push to secure a minor but verified percentage of material — specifically, the first 10-ton total matrix/debris tranche — to prove the "Dust-for-Dollars" micro-tranche mechanism functions perfectly under real-world conditions. The IAEA certifies this first physical tranche, triggering the \$200 million per-ton escrow release and demonstrating the full financial verification pipeline under operational stress.

Table V-6 Phase I Implementation Timeline

Day	Activity	Political Milestone
Day 1	CANDOR advance team arrives in Iran; equipment staging begins at designated logistics hub	Operational presence confirmed
Day 3	IAEA core drilling team initiates subsurface assay at all three collapsed zones	Technical assessment underway
Day 5	Encrypted Video Network achieves full operational status; live feeds transmit to Washington, Tehran, Beijing, Vienna, and Moscow	The Media Win: "immediate transparency" delivered
Day 7	Preliminary assay results confirm Nuclear Dust presence and distribution; AEOI issues initial shift authorizations	First hard data confirms target viability
Day 10	CANDOR installs and tests dual-key digital infrastructure; first joint key-holding drill occurs	Dual-key sovereignty architecture live
Day 14	IAEA issues initial certification confirming 440.9 kg HEU distribution; Swiss facilitator releases \$500M initial tranche	The First Hard Metric: reciprocity proven
Day 15	Active extraction commences in Zone 1 under dual-key authorization	Physical extraction begins
Day 22	Zone 1 extraction reaches 5-ton matrix milestone	Operational pipeline confirmed
Day 30	First 10-ton total matrix tranche achieves IAEA certification; \$200M per-ton release triggers; Phase I report issued to all Parties	The Compressed Demonstration: "Dust-for-Dollars" proven

7.2 Phase II: The Sustainable Operational Grind (Days 31–120)

Once the political narrative stabilizes by the Day 30 milestones, the workforce transitions to a safer, sustainable operational rhythm.

The 12-Hour Shift Cap

The framework enforces a strict maximum of 12-hour shift rotations. Operators handling heavy excavation equipment under radiological containment conditions do not burn out or commit fatal tracking errors. This cap is non-negotiable and operates as an automatic constraint on all scheduling

software. No shift exceeds 12 hours. No operator works consecutive shifts without an 8-hour rest period. Violation of the 12-Hour Shift Cap constitutes a Tier 2 Material Disruption.

Days 31–75 — Primary Extraction Across Zone 1 and Zone 2

Focus shifts entirely to primary extraction. Zone 1 operations complete and transition into Zone 2. CANDOR targets 50% cumulative recovery (220.5 kg HEU equivalent) by Day 75. Engineering teams perform equipment maintenance and calibration during scheduled shift changes without delaying the broader political framework. The 12-Hour Shift Cap ensures operator safety and equipment reliability throughout this intensive period.

Days 76–120 — Excavation Transitions to Zone 3

Excavation transitions to Zone 3, building up to the 95% cumulative recovery mark (418.9 kg HEU equivalent). This 45-day block gives engineering teams the breathing room to handle equipment maintenance and calibration without delaying the broader political framework. By Day 120, CANDOR achieves 95% cumulative recovery, leaving only the final fractional dust for Phase III consolidation.

Table V-7 Phase II Extraction Schedule

Period	Target	Cumulative Recovery
Days 31–45	Zone 1 completion and Zone 2 initiation	35% (154.3 kg)
Days 46–75	Zone 2 primary extraction	50% (220.5 kg)
Days 76–90	Zone 3 initiation; equipment calibration window	65% (286.6 kg)
Days 91–105	Zone 3 primary extraction	80% (352.7 kg)
Days 106–120	Final extraction push across all zones	95% (418.9 kg)

7.3 Phase III: The Orderly Wrap-Up & Exit (Days 121–180)

Days 121–150 — The Final Clean Sweep

Final extraction of the remaining 4.7% fractional dust (21.7 kg HEU equivalent) proceeds with full consolidation into IAEA-certified containers. The IAEA conducts final mass spectrometry and certifies 99.7% recovery. All material is consolidated, sealed, and prepared for transport. This phase operates under the same 12-Hour Shift Cap, with crews handling the most precise and delicate extraction operations of the entire mission.

Days 151–180 — The Logistics Pipeline

Secure convoy preparation and escorted transit to neutral-country storage (Oman or Turkiye, by mutual agreement) proceeds on the following schedule: container convoy preparation and security escort coordination with Iranian police and CANDOR security (Days 151–160); escorted transport of containers to the neutral-country storage facility (Days 161–175); final storage confirmation, IAEA chain-of-custody documentation completion, release of the remaining escrow balance, and disbursement of the \$1 billion completion bonus (Days 176–180).

Table V-8 Phase III Implementation Timeline

Period	Activity	Deliverable
Days 121–135	Final 4.7% recovery extraction across all zones	99.7% cumulative recovery achieved
Days 136–140	All material consolidated into IAEA-certified containers; tamper-evident seals applied	Full containerization complete
Days 141–150	Final IAEA mass spectrometry and certification of 99.7% recovery	IAEA final certification issued
Days 151–160	Container convoy preparation; security escort coordination with Iranian police and CANDOR security	Convoy ready for transit
Days 161–175	Escorted transport of containers to neutral-country storage facility (Oman or Turkiye, by mutual agreement)	Material in secure storage
Days 176–180	Final storage confirmation; IAEA chain-of-custody documentation completed; remaining escrow balance and \$1B completion bonus released	Mission complete; all funds disbursed

7.4 Extension Protocol

If CANDOR determines before Day 150 that the 180-day timeline is insufficient to achieve 99.7% recovery, CANDOR submits a written extension request to all Parties. The extension requires mutual written agreement; the framework limits extension to a single 60-day period. The framework permits no extension beyond 240 total days. If CANDOR does not achieve 99.7% recovery by Day 240, Article 5 (Emergency Break Points) governs automatically.

Article 8: Relationship to Existing Frameworks

8.1 Conjunction with Annex U

This Annex V operates in conjunction with Annex U (Technical Demarcation and Stabilization Addendum). Any territorial breach affecting CANDOR operational sites or logistics corridors triggers the Automatic Breach mechanism under Annex U, Article 3. The Shared Digital Registry under Annex U, Article 5 records all Annex V telemetry data, financial tranche releases, and mass balance certifications.

Any event classified as a Tier 2 or Tier 3 breach under this Annex V simultaneously triggers the corresponding breach mechanism under Annex U, ensuring synchronized penalty application across both frameworks.

8.2 Conjunction with Annex M

This Annex V recognizes the territorial baselines established in Annex M (Territorial Baseline and Ceasefire Demarcation Protocol). The Current Operational Footprint as defined in Annex M, Article 1.1 contains all CANDOR operational sites. No CANDOR activity alters the territorial baselines established in Annex M.

8.3 Integration with the Parent Framework

This Annex V integrates with the parent framework (The Final Steps Accord, Phase I) through the following linkages:

- **Article 1 (HEU Custody Exchange Protocol):** Annex V provides the physical extraction mechanism for the HEU addressed in the parent framework's custody architecture
- **Article 8 (Neutral Escrow):** The Swiss facilitator administers the \$12 billion escrow through the same architecture established in the parent framework
- **Article 10 (Regional Stability Reconstruction Credits):** The \$500 million humanitarian portion of the completion bonus flows through the UN OCHA mechanism established under Article 10
- **Article 12 (Distributed Guarantor Matrix):** China, Russia, the EU, and Oman serve as guarantors for Annex V operations, consistent with their guarantor roles in the parent framework
- **Article 13 (Human-in-the-Loop Governance & Step-Back Architecture):** This Annex V incorporates by reference the parent framework's Article 13 Three-Tier Breach & Step-Back system and calibrates it for Nuclear Dust operations in Article 4

8.4 No Amendment of Parent Framework

This Annex V does not amend the parent framework. It is a modular, self-contained addendum that addresses a specific operational gap (Nuclear Dust remediation) not fully covered by existing annexes. This Annex V supplements, not replaces, all provisions of the parent framework. All parent framework provisions remain operative.

Article 9: Incorporation by Reference

This Annex V is incorporable into any existing or future agreement through the following sentence:

"The Parties adopt Annex V (Sub-Surface Remediation & Joint Sovereignty Protocol) as an operative provision of this Agreement. In the event of any inconsistency, Annex V governs with respect to Nuclear Dust remediation, CANDOR governance, dual-key sovereignty, financial micro-tranche sequencing, three-tier breach and step-back protocols, earnback architecture, verification telemetry, and emergency break-point protocols."

This Annex V needs no amendment of the parent framework. This Annex V enters into force upon signature by both Parties or upon provisional application by exchange of diplomatic notes.

Article 10: Provisional Application

10.1 Provisional Application Period

The Parties apply this Annex V provisionally for 90 days, pending formal incorporation. Exchange of diplomatic notes between the United States and the Islamic Republic of Iran effects provisional application, with notification to the Joint Secretariat.

10.2 Operative Binding During Provisional Application

During provisional application, all provisions of this Annex V are operative and binding, including: the full CANDOR extraction mandate; the dual-key sovereignty mechanism; the Dust-for-Dollars micro-tranche schedule; the Three-Tier Breach & Step-Back Architecture; the Earnback Protocol; and the emergency break-point protocols.

10.3 Termination of Provisional Application

Either Party terminates provisional application with 14 days' written notice to the Joint Secretariat and all guarantor states. Termination of provisional application does not affect:

- Extraction work completed during provisional application
- Financial tranches already released and paid
- IAEA certifications already issued
- Breaches that occurred during provisional application

Note: The 14-day notice period allows for orderly transition without leaving active operations ungoverned. The CANDOR technical leadership executes an orderly wind-down of operations during the notice period, prioritizing the security of all extracted material and the maintenance of seal integrity at all sites.

Signature Block

**For the Government of the Islamic Republic of Iran
(as host sovereign and affected Party):**

Name: _____

Title: _____

Date: _____

**For the United States of America
(as endorsing Party):**

Name: _____

Title: _____

Date: _____

**For the People's Republic of China
(as CANDOR Co-Lead):**

Name: _____

Title: _____

Date: _____

**For the International Atomic Energy Agency
(as Verification Co-Lead):**

Name: _____

Title: _____

Date: _____

**Witnessed by the Joint Secretariat (Muscat), the Russian Federation (Escrow Guarantor), and the
European Union (E3+):**

Signature: _____

Date: _____

Annex V — Sub-Surface Remediation & Joint Sovereignty Protocol — Version 2.5

All obligations execute in active present tense — Expires upon completion of remediation or invocation of Article 5 entombment protocol

Step-Back/Earnback Architecture integrated from The Final Steps Accord Phase I v4.0, Article 13

Immutable Present-Tense Matrix V2.5 governs all clause construction